

Aflevering 1 - Mikkel Moth Billing

Opgave 1.a. Vis at $\langle x, y \rangle = \sum_{n=-\infty}^{\infty} x_n \overline{y_n}$ er et indre produkt på l^2 , hvor $x = (x_n)_{n \in \mathbb{Z}}$ og $y = (y_n)_{n \in \mathbb{Z}}$.

Hvis $x, y \in l^2$, så vis at $\langle x, y \rangle$ giver mening og er et complex-tal.

To show that we have an inner product, we check with Definition (0,1):
 $x, y \in \mathbb{C}^n$

$$\bullet \langle x, x \rangle = \sum_{n=-\infty}^{\infty} x_n \overline{x_n} = \sum_{n=-\infty}^{\infty} (a_n + ib_n) \cdot (a_n - ib_n) = \sum_{n=-\infty}^{\infty} a_n^2 + b_n^2$$

We see that this sum is bigger than 0 if x is not 0

$$\bullet \langle \overline{x}, y \rangle = \sum_{n=-\infty}^{\infty} \overline{x_n} \overline{y_n} = \sum_{n=-\infty}^{\infty} y_n \overline{x_n} = \langle y, x \rangle$$

$$\bullet \langle c \cdot x, y \rangle = \sum_{n=-\infty}^{\infty} c x_n \overline{y_n} = c \cdot \sum_{n=-\infty}^{\infty} x_n \overline{y_n} = c \langle x, y \rangle$$

Since c is the same for all parts of the sum, we can move it outside

$$\begin{aligned}
 * \quad u \in \mathbb{C}^n, \quad \langle u+x, y \rangle &= \sum_{n=-\infty}^{\infty} (u+x)_n \cdot \bar{y}_n = \sum_{n=-\infty}^{\infty} u_n \bar{y}_n + \sum_{n=-\infty}^{\infty} x_n \bar{y}_n \\
 &= \sum_{n=-\infty}^{\infty} u_n \bar{y}_n + \sum_{n=-\infty}^{\infty} x_n \bar{y}_n = \langle u, y \rangle + \langle x, y \rangle
 \end{aligned}$$

So $\langle x, y \rangle$ is a complex inner product

Opgave 1.b. For $n > 0$ lad

$$f_n(t) = \begin{cases} 1 & (0 \leq t \leq \frac{1}{n}) \\ 0 & (\frac{1}{n} < t \leq 1) \end{cases}$$

Vis at $f_n \rightarrow 0$ i $L^2([0, 1])$, mens f_n ikke konvergerer uniformt til 0 på $[0, 1]$.

To show the first part
we use def 0.7

$$\|f_n\|_{L^2}^2 = \int_0^{\frac{1}{n}} 1^2 dt + \int_{\frac{1}{n}}^1 0^2 dt = t \Big|_0^{\frac{1}{n}} = \frac{1}{n}$$

$$\text{as } n \rightarrow \infty \quad f_n \rightarrow 0$$

f_n does not converge uniformly
because some part is always
equal to 1 as $n \rightarrow \infty$

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Opgave 1.c. Find den ortogonale projektion i $L^2([-\pi, \pi])$ af funktionen $f(x) = x^3$ på underrummet $V_n \subseteq L^2([-\pi, \pi])$ udspændt af

$$\left\{ \frac{1}{\sqrt{2\pi}}, \frac{\cos(jx)}{\sqrt{\pi}}, \frac{\sin(jx)}{\sqrt{\pi}} : j = 1, \dots, n \right\}$$

for $n = 1, n = 2$ og $n = 3$. Plot projektionerne sammen med f ved hjælp af CAS værktøj.

V_n is a orthonormal basis, so we use Theorem 0.21:

$n=1$:

$$\langle f, v_1 \rangle \cdot v_1 = \int_{-\pi}^{\pi} x^3 \frac{1}{\sqrt{2\pi}} dx \cdot \frac{1}{\sqrt{2\pi}} = 0$$

$$0 + \langle f, v_2 \rangle \cdot v_2 = \int_{-\pi}^{\pi} x^3 \frac{\cos(x)}{\sqrt{\pi}} dx \cdot \frac{\cos(x)}{\sqrt{\pi}} = 0$$

$$\begin{aligned} p_1 &= \langle f, v_3 \rangle v_3 = \int_{-\pi}^{\pi} x^3 \frac{\sin(x)}{\sqrt{\pi}} dx \cdot \frac{\sin(x)}{\sqrt{\pi}} \\ &= 2 \cdot (\pi^2 - 6) \cdot \sin(x) \end{aligned}$$

The first 2 parts will be 0 for all j :

$n=2$

$$p_2 = \langle f, v_3 \rangle \cdot v_3 = \int_{-\pi}^{\pi} x^3 \frac{\sin(2x)}{\sqrt{\pi}} dx \cdot \frac{\sin(2x)}{\sqrt{\pi}}$$

$$= (3 - 2\pi^2) \cdot \frac{\sin(2x)}{2} \quad 3/4$$

$$n=3$$

$$p_3 = \langle f, v_3 \rangle v_3 = \int_{-\pi}^{\pi} x^3 \frac{\sin(3x)}{\sqrt{\pi}} dx \frac{\sin(3x)}{\sqrt{\pi}}$$

$$= (6\pi^2 - 4) \cdot \frac{\sin(3x)}{9}$$

